NAMILYANGO COLLEGE

A 'LEVEL PRINCIPAL MATHEMATICS SEMINAR 2023

PART 1 – PURE MATHEMATICS

A) ALGEBRA

1. If $a^2 + b^2 = 23ab$, show that $loga + logb = 2log\left(\frac{a+b}{5}\right)$.

2. Given that one of the roots of the equation $x^2 + px + q = 0$ is twice the other root, show that $2p^2 = 9q$. Hence find the value of k if the equation $x^2 - 2(k+2) + (k^2 + 3k + 2) = 2$ has one root twice the other.

3. Prove by induction that $\sum_{r=1}^{n}(r+1)2^{r-1} = n\left(2^n\right)$

4. (a)Express $\left(\frac{2+i}{1-i}\right)^6$ in polar form.

   (b) Solve the equation $z^2 - (5+4i)z + 10i = 0$.

5. a) i) Find the coefficient of x^{-6} in the expansion of $\left(x^2 + \frac{1}{x^2}\right)^{12}$.

   ii) Find the constant in the expansion of $\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^6$.

   b) Use binomial theorem to evaluate;

   i) $(98)^4$ ii) $3\sqrt{128}$ (to 4dp).

6. a) Given that $\left|\frac{z-1}{z+1}\right| = 2$, find the cartesian equation of the locus of Z where Z is a complex number and represent the locus on an argand diagram.

   Shade the region for which the inequalities $\left|\frac{z-1}{z+1}\right| > 2$ and $0 < \arg < \frac{3\pi}{4}$.

   b) i) Given that x and y are real, find the values of x and y which satisfy the equation $\frac{2y+4i}{2x+y} - \frac{y}{x-i} = 0$.

   ii) Given that $z = x + iy$, where x and y are real, show that when $im\left(\frac{z+i}{z+2}\right) = 0$, the point (x, y) lies on a straight line and when $\text{Re}\left(\frac{z+i}{z+2}\right) = 0$, the point (x, y) lies on a circle. Determine the center and radius.

c) Without a calculator, simplify $\dfrac{\left(\cos\frac{2\pi}{5}+i\sin\frac{2\pi}{5}\right)^8}{\left(\cos\frac{3\pi}{5}-i\sin\frac{3\pi}{5}\right)^3}$.

7. Express $z_1=\dfrac{7+4i}{3-2i}$ in the form $p+qi$ where p and q are real numbers.

   Sketch in an argand diagram the locus of points representing complex number Z such that $|z-z_1|=\sqrt{5}$.

   Find the least value and the greatest value of $|z|$object to this condition.

8. a) Determine the interval in which $\dfrac{2x}{x-1}+\dfrac{x-5}{x-2}>3$ lies.

   b) Given that $y(x-1)=x^2+3$, where x is real, show that y cannot take any value between -2 and 6.

   c) Find the asymptotes of the curve $y=\dfrac{x^2+3}{x-1}$ and sketch the curve, showing clearly the coordinates of turning points.

9. a) If ${}^nC_{12}={}^nC_8$, find ${}^{22}C_n$.

   b) Find the number of arrangements that can be made from the word *DOGMATIC* if,

   i) There are no restrictions. ii) Provided the letters D,O,G are not to be separated.

   c) find the number of ways in which a soccer team consisting of 11players can be selected from a squad of 16 players.

   i) how many ways would be if the captain must be included.

   ii) how many ways would be if the captain is left out injured.

B) TRIGONOMETRY

1. a) Solve for x in the equation $cos^{-1}2x-cos^{-1}x=\frac{\pi}{3}$.

   b) Solve the equation $sin^{-1}x-cos^{-1}x=2x^2-1$.

2. Solve the equations;
   a) $5tan\theta+sec\theta=0$ for values of θ between 0^0 and 360^0
   b) $8cos^4x-cos^2x+3=0$ for x in the range $0^0\le x\le 360^0$

3. a) Show that $12\sin^2 x\cos^2 x = \frac{3}{2}(1-\cos 4x)$.

b) Solve $4x = \cos 3x$ for x from 0 to 180^0 inclusive.

c) Given that $\cos 45^0 = \frac{1}{\sqrt{2}}$, show without using a calculator that $\sin\left(292\tfrac{1}{2}\right)^0 = \frac{1}{2}\sqrt{\left(2+\sqrt{2}\right)}$.

4. a) Express $\dfrac{\sin 16\theta\cos 2\theta - \cos 6\theta\sin 12\theta}{\cos 4\theta\cos 2\theta - \sin 6\theta\sin 8\theta}$ as $\tan k\theta$ where k is a constant. Hence state the value of k.

b) The sides BC, CA and AB of a triangle ABC are of length $(x+y)$, x and $x-y$ respectively. Show that $\cos A = \dfrac{x-4y}{2(x-y)}$.

c) Show that in any triangle ABC, if $2s = a+b+c$, then $1-\tan\frac{A}{2}\tan\frac{B}{2} = \frac{c}{5}$.

C) VECTORS

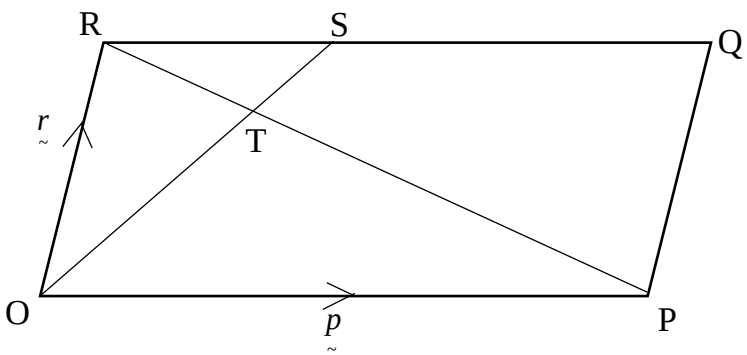
1. (a) Given the line and plane $r = t\hat{\imath} - 2t\hat{\jmath} + (1-t)\hat{k}$ and $3x + 4y - z = 1$. Find;

   i) The position vector of their point of intersection.

   ii) The Acute Angle between the line and the plane.

   b) Find the unit vector which is perpendicular to both the vectors $\hat{\imath} + 3\hat{\jmath} - 5\hat{k}$ and $2\hat{\imath} - \hat{k}$.

2. In the figure below, $OPQR$ is a parallelogram with $\mathbf{OP} = \underset{\sim}{p}$, $\mathbf{OR} = \underset{\sim}{r}$. S is the mid-point of QR and OS meets PR at T.



Given that $OT = mOS$ and $PT = nPR$;

i) Find in terms of $\underset{\sim}{p}$ and $\underset{\sim}{r}$, the vectors, $\mathbf{OT}$, $\mathbf{PT}$ and $\mathbf{RT}$

ii) $TR : PT = 1 : 2$

3. a) the equations of the two planes are

$$\underset{\sim}{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + t\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + s\begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \text{ and } \underset{\sim}{r} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \mu\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} + s\begin{pmatrix} -2 \\ 4 \\ 3 \end{pmatrix}$$

i) Determine the cosine of the acute angle between the planes.

ii) Find the equation of the perpendicular distance of the point $P(1,5,1)$ from the line L.

b) Determine the vector equation of the plane containing the lines,

$$\underset{\sim}{r} = (\lambda + 1)\hat{i} - 2\hat{j} + 3\hat{k} \text{ and } \underset{\sim}{r} = 3t\hat{i} + (2t - 4)\hat{j} + (t + 2)\hat{k}$$

4. a) A line L_1 passes through the point (-4, 7, 5) and is parallel to the vector $3\hat{i} - 2\hat{j} + 4\hat{k}$. Find,

i) The point of intersection of L_1 and the line $\frac{x}{2} = \frac{y+5}{1} = \frac{z+11}{8}$.

ii) The equation (cartesian) of the plane containing L_1 and the origin.

b) Determine the equation of the plane that passes through the points $P(1,0,2)$ and $Q(3,-1,1)$ parallel to the line with the vector equation $\underset{\sim}{r} = 3\hat{i} + (2t - 1)\hat{j} + (5 - t)\hat{k}$.

D) GEOMETRY

1. a) i) Obtain the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(acos\theta, bsin\theta)$.

ii) Find the eccentricity of the ellipse $x^2 + 16y^2 = 25$.

b) Show that the point $P(ct, {}^{c}/_{t})$ lies on the rectangular hyperbola $xy = c^2$, determine the equation of the tangent to the hyperbola at point P. Deduce also the locus of the point of intersection of the foot of the perpendicular from the origin to the tangent.

2. a) Find the Eccentricity, Foci and directrices of the following ellipses;

i) $4x^2 + y^2 = 1$.

ii) $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

iii) $16x^2 + 25y^2 = 100$.

b) A point moves so that its distance from $A(3,2)$ is half its distance from the line $2x+3y=1$. Why is the locus an ellipse? Find the equation of the major axis.

c) Find the equation of the tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$at the end of the latus rectum which lies in the first quadrant.

3. Show that the graph of $y=\frac{x^2-1}{x^2+4}$ has no real stationary points and sketch the graph.
4. a) A circle C has equation $x^2+y^2-4x+2y=40$. Find the equations of the circles with center $(3,1)$, which touch the circle C. Find the coordinates of the point of contact.
   b) Given that the circles and intersect at P and Q, find the equation of the circle which pass through P, Q and
   i) The point (1,1)
   ii) The origin.
   iii) Touches the x-axis.

E) ANALYSIS

1. a) Given that $P = x(x^2+4)^{\frac{1}{2}}$, find $\frac{dP}{dt}$ when $x = 2$, and $\frac{dx}{dt} = 1$.

   b) Given that $x^y = e^{x-y}$, show that $\frac{dy}{dx} = \frac{lnx}{(lnx+1)^2}$.
2. The area bounded by the curve $y = 15x(k-x)$ and the x-axis is rotated through 2π radians about the x-axis. If the volume of the solid generated is 240π cubic units, find the value of k.
3. Evaluate $\int_0^{\frac{\pi}{2}} x^2 sinxcosxdx$
4. a) Using Maclaurin's theorem, show that $cosx = 1-\frac{x^2}{2!}+\frac{x^4}{4!}+---$. Hence evaluate $cos11.2^{\circ}$. to three places of decimal.

   b) Determine the first 4 terms of the Maclaurin's expansion of $tan^{-1}x$, hence evaluate the integral $\int_0^1 tan^{-1}xdx$.
5. (a) i) Differentiate x^{lnx} with respect to x.

   ii) Given that $cos\theta = \frac{x}{x+1}$,$sin\theta = \frac{y}{y+1}$, find $\frac{dy}{dx}$.

   b) Show that the greatest volume of a right circular cone that can be inscribed in a sphere of radius a is $\frac{32\pi a^3}{81}$ cubic units.

6. Integrate the following

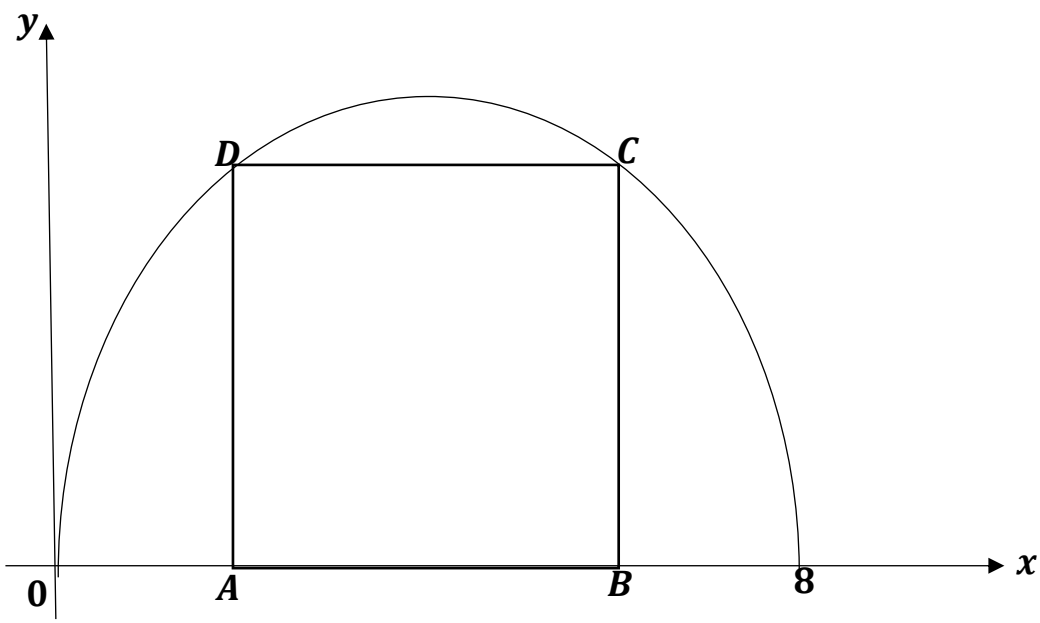
(a) $\int 2^{\sqrt{x-2}}\,dx$

(b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{4\cos x}{3+\cos^2 x}\,dx$

(c) $\int \tan^3 x dx$

(d) Evaluate $\int_2^4 \frac{dx}{\sqrt{(x^2-4)}}$

7.



The above figure shows part of the parabola $y = 8x - x^2$ with a rectangle $ABCD$ which fits between the curve and the x axis. Taking $AB = 2x$.

a) Show that $OB = x + 4$
b) Prove that the area of $ABCD$ is $32x - 2x^3$ sq units. Hence find the value of x which makes the area a maximum and state the maximum area.

8. In 1977 the world population was 4.3 billion persons and it was increasing at the rate of a quarter of a million persons each day (if the birth rate and the death rates were constants), then answer the following questions;
   a) Determine what the annual growth rate was.
   b) Determine how long it will take the world's population to double
   c) Determine what would be the world's population by the year 2000
   d) Determine when will the world's population be 50billion

9. Solve the differential equation; $\sqrt{(2x^2-7)}\frac{dy}{dx}=4xy^2$

10. a) Find $\frac{dy}{dx}$ if,

i) $y=\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$.

ii) $y=\sqrt{\left(\frac{2x+1}{2x-1}\right)}$.

b) find $\frac{d^2y}{dx^2}$ in terms of the parameter t when,

i) $x=\cos^3 t,\quad y=\sin^3 t$ ii) $x=t^2-4,\quad y=t^3-4t$

11. a) Differentiate from first principles, x^3-4x^2+1.

b) Investigate the stationary values (points) given that;

i) $x^3+y^3-3xy=0$.

ii) $y=\frac{x^3}{1+x^2}$.

c) Solve the following differential equations.

i) $x\frac{dy}{dx}-3=2y+\frac{dy}{dx}$.

ii) $(1-2x)e^y\frac{dy}{dx}-2e^y=\sec^2 x$.

iii) $3y+(x-2)\frac{dy}{dx}=\frac{2}{x-2}$.

12. Evaluate

i) $\int_0^{\frac{\pi}{2}}\frac{\sin^3\theta}{2-\sin^2\theta}\,d\theta$.

ii) $\int_0^{\sqrt{2}} xe^{-x^2}\,dx$.

iii) $\int\frac{\cos\theta}{1-\sec\theta}\,d\theta$.

13. a) Solve the differential equations,

i) $x\frac{dy}{dx}-y=x^2\ln x$.

ii) $\frac{dy}{dx}+y\cot x=\cos 3x$ for $y\left(\frac{\pi}{6}\right)=1$.

b) During a fermentation process the rate of decomposition of substance Y at any instant is given by $\frac{dy}{dx}=-2xy$ where y kg is the amount of y left and x kg is the amount of active fermenter in Y. if the value of x at time t is $\frac{1}{2(1+t)^2}$;

i) Express $\frac{dy}{dt}$ in terms of y and t. Hence express y in terms of t.

ii) If $y=40$ when $x=10$, find y when $t=10$.

iii) Show that the amount Y remaining when the fermentation has stopped is approximately 14.6 kg.

14. Find the value of each of the following,

a) $\int_{\frac{1}{2}}^{1}\frac{x-2}{(x+2)^3(x-6)^3}dx$.

b) $\int\frac{\text{Sec}^2\left(\sqrt{x}\right)}{\sqrt{x}}dx$.

c) $\int\frac{4x^2}{\sqrt{(1-x^6)}}dx$.

15. a) Find the area of the region of the xy plane defined by the inequalities, $y^2\le 1-x$ and $y\le x+1$.

b) The volume of the cartesian face is equivalent to that generated when the area enclosed by $x=\frac{y^3}{64}+1$, the y-axis and $y=8$ is rotated through 2 right angles about y-axis, calculate its volume.

16. Express $\frac{2(x+2)^2}{x^2(x^2+4)}$ into partial fractions, hence evaluate $\int_1^2\frac{2(x+2)^2}{x^2(x^2+4)}dx$.

17. a) Find in form $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the equation of the ellipse with eccentricity $\frac{1}{2}$ and foci $(\pm 2,0)$.

b) Use the locus definition of the ellipse with eccentricity $\frac{2}{3}$, focus (2,1) and directrix $x=-\frac{1}{2}$.

c) Determine the equation of the parabola whose focus is the point $B(3,4)$ and the directrix is the line $x+y=1$.

Determine the coordinates of the center and the foci of the ellipse $\frac{(x-1)^2}{9}+\frac{(y-2)^2}{5}=1$. Also determine the equation of the directrices.

18. a) Evaluate $\int_0^a x^3\left(a^2-x^2\right)^{\frac{3}{2}}dx$.

b) Evaluate $\int_0^{\frac{\pi}{2}} x\sin^2 x\,dx$.

c) Evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\cos^2 x\,dx$.

d) If $\int_0^{\frac{\pi}{2}}(x\sin x)\,dx=\int_0^2\left(ax^2+2x\right)dx$ find the value of a.

e) Evaluate $\int_1^2 \frac{dx}{x^2\sqrt{x-1}}dx$.

f) Evaluate $\int_1^2 \frac{dx}{x^2\sqrt{5x-1}}dx$.

19. Show that,

a) $\frac{d}{dx}\left(x^2\tan^{-1}(x^2)\right)=2x^3\left(1+x^4\right)^{-1}+2x\tan^{-1}(x)$.

b) If $y^2-2y\sqrt{\left(1+x^2\right)}+x^2=0$, then $\frac{dy}{dx}=\frac{x}{\sqrt{\left(1+x^2\right)}}$.

20. A piece of wire 80 cm in length is cut into 3 parts two of which are bent into equal circles and the third into a square. Find the radius of the circles if the sum of the enclosed areas is a minimum.

PART 2 – APPLIED MATHEMATICS

A) MECHANICS

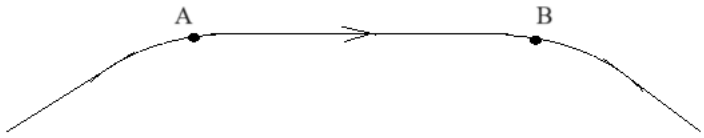
1. a) i) Point O is the origin and Points A, B, C and D have position vectors $\begin{pmatrix}3\\1\end{pmatrix}m$, $\begin{pmatrix}1\\3\end{pmatrix}m$, $\begin{pmatrix}-2\\1\end{pmatrix}m$ and $\begin{pmatrix}-2\\-2\end{pmatrix}m$ respectively. Forces $\begin{pmatrix}3\\3\end{pmatrix}N$, $\begin{pmatrix}4\\-5\end{pmatrix}N$, $\begin{pmatrix}-5\\2\end{pmatrix}N$ and $\begin{pmatrix}2\\3\end{pmatrix}N$ act at points A, B, C, and D respectively. Find the single Force that could replace this system.

   ii) Find where its line of action cuts the horizontal axis.

   iii) A couple of moment $a\,Nm$ acting anti-clockwise and a force $\begin{pmatrix}b\\c\end{pmatrix}$ N, acting through the point which has position vector $\begin{pmatrix}2\\1\end{pmatrix}m$ are now added to the system. If these reduce the system in equilibrium, find a, b and c.

   b) ABCDEF is a regular hexagon. Forces of magnitudes 2, 3, 4, 5, and 6 N act in the directions of AB, AC, AD, AE and AF respectively. If their resultant is a force of R N acting at an angle θ to Ab, find R and θ.

2.



The diagram above shows part of a ring circuit with two bends and a straight-line AB. A car moves out of the bend at A with speed of 12 m/s and accelerated uniformly in the direction shown reaching a top speed of 48 m/s in 6 s.

Find,

a) The acceleration.
b) The distance travelled from A to reach the top speed.

The driver maintains a speed of 48 m/s for 2 s. he then decelerates uniformly at 7.5 m/s^2 until B is reached. Given that the distance from A to B is 408 m, find,

c) The speed of the car at B.
d) The time taken from A to B.
e) Ketch a velocity – time graph for the motion from A to B.

3. A particle is projected from O at time $t = 0$ and performs simple harmonic motion with O as the centre of oscillation. The motion is of amplitude 20 cm and time period 4 seconds. Find
   (i) the speed of projection
   (ii) the speed of the particle when $t = 1.5\ seconds$
   (iii) the value of t when the particle is first at a point 10 cm from O.
   (leave π in your answers where they occur)
4. a) A particle travelling in a straight line with constant acceleration covers distances S_1 and S_2 in the third and fourth seconds of its motion respectively. Show that the initial speed is given by;

$$\left(\frac{7S_1 - 5S_2}{2}\right)$$

   b) The maximum possible acceleration of a body is 1 m/s^2 and the maximum possible retardation is 3 m/s^2. What is the least time in which it can travel 1 km from rest to rest?

5. a) Determine the center of gravity of;
   i) a uniform solid right circular cone base radius r and height h from the vertex.
   ii) a uniform solid hemisphere of radius r from the plane surface.

   b) A solid consists of a uniform right circular cone of density ρ, radius r and height $4r$ mounted on a uniform hemisphere of density σ and radius r, so that the plane faces coincide. Show that the distance of the centre of mass of the whole solid from the common plane face is,

$$\frac{r}{8}\left(\frac{16\rho - 3\sigma}{2\rho + \sigma}\right)$$

   If $\rho = \sigma$ and the solid is suspended freely by a string attached to a point on the rim of the common face, find the inclination of the axis of the cone to the vertical.

6. ABCD is a square lamina of side a, from which a triangle ABE is removed, E being a point in CD distant c from point C. Show that the center of gravity of the remaining lamina is distant $\dfrac{a^2 + ac + c^2}{3(a+c)}$.

Deduce that if this lamina is placed in a vertical plane with CE resting on a horizontal table, equilibrium will not be possible if c is less than the positive root of the equation $2x^2 + 2ax - a^2 = 0$.

7. a) Particles of mass 5m, 4m and 3m kgs are placed at points $(-5, 0)$, $(4,\ 0.5)$ and $(-4, -3)$ respectively. Determine the point where the mass 0f 7m kg should be placed to ensure that the center of gravity of the entire system lies at the origin.

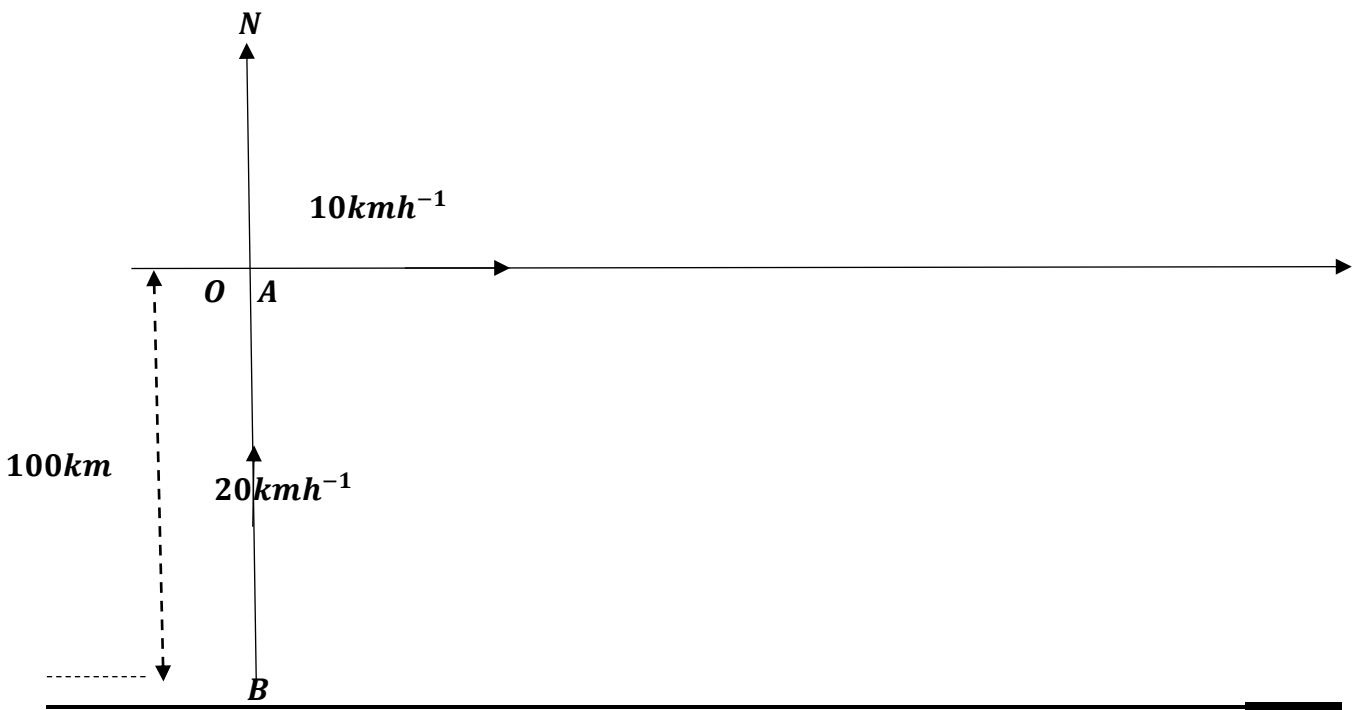
   b) Find the centroid of the volume obtained by rotating completely about the x-axis the area bounded by the curve $y = x^2$, the x-axis and the line $x = 1$.

8. A uniform rod of length l rests in a vertical plane against (and over) a smooth horizontal bar at a height h above the ground, the lower end of the rod being on level ground. Show that if the rod is on the point of slipping when the inclination to the horizontal is θ, then the coefficient of friction between the rod and the ground is; $\dfrac{l\sin 2\theta \sin\theta}{4h - l\sin 2\theta\cos 2\theta}$.

9. A particle of mass m kg is placed on a rough plane inclined at an angle of 30° to the horizontal. Given that the angle of friction is $\lambda > 30^{\circ}$, show that the minimum force required to move the body up the plane is given by, $\frac{1}{2}mg\left(cos\lambda + \sqrt{3}sin\lambda\right)$.

   If this force is 3 times the least force that would cause the body to move down the plane. Show that $\lambda = tan^{-1}\left({}^{2}/_{\sqrt{3}}\right)$.

10. Ship A is at O at noon and sailing due East at $10kmh^{-1}$ as shown in the figure below;



At that time ship B is $100km$ due south of O and is sailing at $20kmh^{-1}$ due north.

a) State the distances in km of A and B from O after t hours
b) Show that the distance Skm between $A\ and\ B$ is then given by;
$S^2 = 500t^2 - 4000t + 10000$
c) Find the value of t for which S^2 is a minimum and hence find the minimum distance.

11. Two vehicles are proceeding, one on each road towards the point of intersection of two roads which meet at an angle of 60^0. If their speeds are 20 and $32kmh^{-1}$ and they are respectively 70 and $40km$ from their cross roads, find;
a) their relative velocity in magnitude
b) their distances from the cross-roads when they are nearest together.

12. A bus is travelling at $20kmh^{-1}$ along a straight road, a man running at $10kmh^{-1}$ along a perpendicular road, sees it when it is $200m$ short of the junction and himself is $150m$, short. Show that the man can never get nearer to the bus than $20\sqrt{5}$.

13. A motor boat sets out at $11{:}00am$ from a position $\begin{pmatrix}-6\\-2\end{pmatrix}$ and travels at a steady speed of $\sqrt{53}$ on a direct course to intercept a ship the ship maintains a steady velocity $\begin{pmatrix}3\\4\end{pmatrix}$ and at $12{:}00\ noon$ is at a position $\begin{pmatrix}3\\-1\end{pmatrix}$ from O. find;
a) The velocity of the motor boat
b) The time of interception
c) The position vector of the boat at interception

14. A particle projected under gravity from a point O passes through the points whose coordinates relative to the horizontal and vertical axes through O are (a, a) and $(2a, a)$ find;
a) The angle of projection
b) Show that the velocity of projection is given by $4V^2 = 13ga$
c) Find the value of V if $a = 4m$ and $g = 10m/s^2$

15. A ball is projected with a velocity Vm/s at an angle θ so that it just clears two walls of equal height bcm which are bcm apart. Prove that;
$$bg = 2Vcos\theta\sqrt{(V^2sin^2\theta - 2bg)}$$
16. A particle P is projected from point O on level ground with speed $50m/s$ at an angle $sin^{-1}\left(\frac{7}{25}\right)$ above the horizontal. Find;
a) The height of P at the point where its horizontal displacement from O is $120m$
b) The speed of P $2s$ after projection

c) The times after projection at which P is moving at an angle of $\tan^{-1}\left(\frac{1}{4}\right)$ to the ground.

17. One end of an elastic string is fixed at A and the other end is attached to a particle heavy enough to stretch the string to double its natural length a. Show that if the particle is dropped from A it will descend a distance $\left(2+\sqrt{3}\right)$ before coming to rest.

18. A mass is suspended from a fixed-point O by an elastic string of a natural length a and when the mass is hanging freely the length of the string is $\frac{5}{3}a$. Show by principal of energy that if the mass to be allowed to fall from rest at O, the greatest length of the string in the subsequent motion is $3a$
Find also the speed with which the mass is moving when it is a distance $2a$ from O

19. A particle of mass $3kg$ is suspended from an elastic string of natural length $0.5m$ and modulus $48N$. If the particle is pulled vertically downwards and then released when the length of the string is $1metre$,find the acceleration at that instant when it is released.

20. a) A light elastic string of natural length $1m$ is fixed at one end and a particle of weight $4N$ is attached to the other end. When the particle hangs freely in equilibrium the length of the string is $1.4m$. the string is now held at an angle θ to the vertical by a horizontal force of $3N$ acting on the particle. Find the value of θ and the new length of the string.

b) One end of a light elastic string of natural length $6m$ and modulus $48N$ is attached to a fixed-point A on a smooth horizontal surface. A body of mass $2kg$ is attached to the other end of the string and is held at rest at a point B on the surface which is at a distance of $9m$ from A. the body is released from rest. Find the time that elapses before the body reaches A.

21. a) A particle of mass $2kg$ is attached to one end of an elastic string of natural length $1m$ whose other end is fixed to a point A on a smooth horizontal plane. The particle is pulled across the plane to a point C where $AC = 1.5m$ and is released from rest AC. B is the point on AC such that $AB = 1m$. If the modulus of the string is $10N$; Show that;
i) From C to B the particle performing simple harmonic motion with centre B

Determine;

ii) The time taken to travel from B to C

iii) the speed at B
iv) the time taken when the particle is moving at constant frequency.

b) A particle moving with simple harmonic motion performs 15 complete oscillations with amplitude $3m$ every minute. Find;
i) the maximum acceleration
ii) the maximum speed
iii) the average speed
iv) the speed when the particle is $2m$ from the central position
v) the particle's position $0.4s$ after leaving the central position.

B) NUMERICAL METHODS

1. Given that $f(x) = x^4 - x - 10,$

(i) locate the ranges where the two real roots of f(x) = 0 lie

(ii) use linear interpolation to find the largest root in (i) above correct to 4 S·f

2. An equation has two iterative formulae $x_{n+1} = 3 - \frac{1}{x_n}$ and $x_{n+1} = \frac{1}{3 - x_n}$.

(i) Without iterating determine with a reason which formula is suitable when $x_o = 2 \cdot 5.$ Hence find the root correct to 3 d·p

(ii) Find the equation that leads to the two iterative formulae

3. (i) By drawing graphs of $y = e^x$ and $y = 4sinx$ on the same axes, find each of the two roots of the equation $e^x - 4sinx = 0$ for $0 \le x \le 1 \cdot 5$ using intervals of 0·3

(ii) Use Newton Raphson method to find the largest root of the equation in (i) above correct to 3 decimal places

4. (i) Show that the sixth root of a number A is given by the sequence:

$$x_{n+1} = \frac{5}{6}\left(x_n + \frac{A}{5x_n^5}\right), \qquad n = 0, 1, 2, ----$$

(ii) Draw a flowchart that computes and prints such a root after 3 iterations and gives it correct to two decimal places.

(iii) Perform a dry run for your flowchart using A = 155 and $x_o = 2$

5. The income tax of an employee is calculated as follows:

Taxable Income, A (£)	Tax rate (%)
01– 400	10
401– 900	20
Above 900	30

Draw a flowchart that reads the taxable income A, computes and prints the income tax ,T, and the taxable income A of 5 employee

6. Two numbers x and y were estimated with errors Δx and Δy respectively.

(i) Write an expression for the maximum percentage error in $\frac{x}{y}$

(ii) If x = 2·5 and y = 1·25, find the percentage error in $\frac{x}{y}$

7. The respective errors in the measurements x and y are Δx and Δy. Show that the maximum relative error in $\frac{x}{y^2}$ is $\left|\frac{\Delta x}{x}\right| + 2\left|\frac{\Delta y}{y}\right|$. Hence if $x = 81 \pm 0 \cdot 405$ and $y = 6 \pm 0 \cdot 15$, find the limits within which the exact value of $\frac{x}{y^2}$ lies

8. Find the limits within which the exact value of $\frac{2 \cdot 54 - 14 \cdot 8}{3 \cdot 7}$ lies

9. (i) Use the trapezium rule with six ordinates to estimate $\int_2^3 \left(\frac{4}{x} + \cos x\right) dx$ correct to 4 decimal places

(ii) Find the error in your estimation in (i) above and suggest how it may be reduced.

10. Use the fact that f(1·15) = 1·32 and $f^{-1}(1 \cdot 26) = 1 \cdot 25$ to find the value of $f^{-1}(1 \cdot 22)$ by linear interpolation correct to 2 d·p

C) STATISTICS AND PROBABILITY

1. Two samples of size 80 and 20 respectively have mean 45 and 50 and standard deviation 6 and 9. Find the combined mean and standard deviation

2. The table below shows the marks of 50 students in a test

Marks	< 20	< 30	< 35	< 40	< 50	< 60
No of students	4	20	32	42	48	50

(a) Construct a frequency distribution table

(b) Calculate the:

(i) mean mark

(ii) number of students who passed, if the pass mark was 38

(c) Draw a histogram for the data and use it to estimate the modal mark

3. In a normal distribution 87·7% of the items are over 30 and 67% are under 70. Find the:

(i) mean and standard deviation of the distribution

(ii) probability that the distribution lies between 40 and 52

(iii) limits within which the central 90% of the distribution lies

(iv) probability that the mean of 16 items selected at random exceeds 51

4. Use the fact that X~ Bin(100 , 0·5), to find:

(i) P(X = 40) (ii) $P(X \leq 53)$ (iii) $P(X \geq 45)$ (iv) P(X > 60)

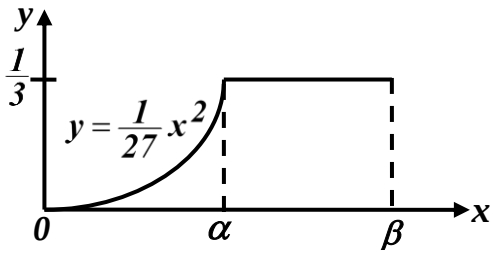
(v) P(X < 48) (vi) $P(43 \leq X \leq 55)$ (vii) $P(41 \leq X < 54)$ (viii) $P(47 < X < 56)$

5. A random sample of 100 bags taken from a normal population had the following weights x in kg: $\sum x = 760$ and $\sum x^2 = 7360.$ Calculate the:

(i) unbiased estimate for the population variance

(ii) 95% confidence interval for the population mean

6. The p·d·f of a continuous r·v X is distributed as follows:



Find:

(i) the values of α and β

(ii) the equations of the p·d·f

(iii) $P(2 \le X \le 4)$

(iv) the mean and median of X

(v) the distribution function of X

7. A continuous r·v X ~ U(α , β) with lower quartile 5 and upper quartile 9.

(a) Write down the p·d·f of the distribution and sketch it

(b) Find: (i) the values of α and β (ii) $P(6 \le X \le 7)$

(iii) the probability that X lies within one standard deviation of the mean

8. A r·v X has the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & , \quad x \le 0 \\ \frac{1}{3}x^2 & , \quad 0 < x \le 1 \\ \frac{1}{6}(\beta x - x^2 + \lambda) & , \quad 1 < x \le 3 \\ 1 & , \quad x > 3 \end{cases}$$

Find:

(i) the values of β and λ, hence sketch F(x)

(ii) the median of X

(iii) $P(0 \cdot 6 \le X \le 2 \cdot 7)$

(iv) the mean of X

9. A discrete r·v X takes the values 1, 2, 3 or 4 and its cumulative distribution function is as follows:

$$F(x) = 1 - \left(1 - \frac{x}{4}\right)^{x} \quad , \quad x = 1, 2, 3, 4$$

Find:

(i) the probability distribution of X and sketch it.

(ii) P(X > m), where m is the median

(iii) mode, mean and variance of X

10. Three pens are drawn at random without replacement from a box containing 4 blue and 5 red pens. If X is the number of red pens drawn, find the:

(i) probability distribution of X

(ii) mode, mean, variance and median of X

11. A biased coin that shows a head and a tail in the ratio 3:7 respectively is tossed 12 times. Find the:

(i) probability of getting more heads than tails

(ii) most likely number of heads

12. Events A and B are such that $P(A) = \frac{1}{2}$, $P(A/B) = \frac{1}{3}$ and $P(B/A) = \frac{2}{5}$.

(a) State with reasons whether A and B are:

(i) independent events (ii) mutually exclusive events

(b) Find: (i) P(A n B) (ii) P(A U B) (iii) $P(A/\overline{B})$

13. The chance of Tom going to work by bus, by train or on foot is 0·3, 0·45 and 0·25 respectively. If he uses these methods of travel, the chance that he will be late is 0·4, 0·65 and 0·8 respectively. Find the chance that he will:

(i) be late on a particular day

(ii) use a bus given that he is not late

(iii) not go on foot given that he is late

14. The price index of an item in 2018 based on 2017 was 125. Its price index in 2019 based on 2018 was 144. Find its price in 2019 if its price in 2017 was Shs 5,000

15. If events A and B are independent,

(i) show that the events $\overline{A}$ and $\overline{B}$ are also independent

(ii) find the possible values of P(A), if $P(A \cap B) = 0 \cdot 3$ and $P(\overline{A} \cap \overline{B}) = 0 \cdot 125$

END